Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
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Friend (optional)	Anthony	

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea

and a paragraph describing how that paper supports your design concept.
        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider the project below.
        - Create an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the chempy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.

People learning computer engineering often need to convert numbers between binary, decimal, and hexadecimal. Performing conversions can be time consuming and can lead to mistakes.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

Programmable calculators, online number base converters, and programmer calculators already provide number conversion.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

A web based number system converter that converts binary, decimal, and hexadecimal numbers, makes sure the input is correct and explains the conversion process.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

Create a web application that converts numbers between binary, decimal, and hexadecimal while helping users understand the conversion process.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

A number the user inputs and the number system it belongs to.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

Make sure the input is correct, convert the number into decimal integer, convert the integer to the requested number systems.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The converted number or any applicable error message.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Number Number system

Conversion Result
Conversion History

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Input Value
Original number system
Decimal number of input value
Converted number

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

Some difficulties could be making sure the input is correct because people are unpredictable.
Representing hexadecimal characters consistently
Perhaps implementing a practice or learning system.
Design a good website with easy to understand interface

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

Ask explanations of each code the LLM provides, and test conversions with unit tests.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

Build a functional website with a visually appealing interface.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

“I want to make a website that converts binary into decimal numbers or hexadecimal into decimal or binary.”

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

AI or online calculators could be an existing solution to my app.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

The most complex or over-ambitious part of my program will be the interactive practice section, I could make it a single practice question instead of interactive practice which generates questions and maintains scores.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

I have gotten many good ideas to add to the project and increase the depth of the project.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I can ask people in the CISC discord server for help, TAs, or even an LLM.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

First I will try to see what part of the code is broken or causing an error and then I will try to fix that part myself or I will ask an LLM to revise that part of the code according to the error.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

where would i start for brainstorming an idea for a website that i can build

The best place to start is by thinking about a small problem you or someone you know has, then figuring out how a website could solve it. You don't need to come up with a completely original idea! For your CS1 final project, a simple website that solves one problem well is better than a complicated website that tries to do everything.

I looked at your project worksheet, and it asks you to identify a problem, research existing solutions, plan your website's inputs and outputs, and narrow your idea to something you can realistically build in three weeks.

Here's how I'd approach brainstorming.

Step 1: Start with your interests

Think about things you already enjoy, do regularly, or sometimes struggle with.

Here are five areas you could explore:

Category	Problem you could solve	Possible website i
🎓 College life	Students struggle to manage assignments and deadlines.	A homework planner that prioriti

💰 Money	Students want to save money but don't know where it goes.	A spending tracker that categori and calculates totals.
🎮 Hobbies	People struggle to keep track of their progress in a hobby.	A gaming, fitness, or reading pro
🍕 Food	Students don't know what to eat with the ingredients they have.	A meal suggestion tool based or ingredients.
🧠 Productivity	People have trouble staying focused while studying.	A study timer that tracks study s displays statistics.

My advice: Start with something you personally relate to. You'll have a better understanding of the problem, which makes it easier to decide what features your website needs.

Step 2: Look for problems, not features

A common mistake is starting with an idea like "I want to make a website with a map, a graph, and a bunch of buttons."

Instead, ask yourself these three questions:

1. What is annoying? What is something you or your friends find inconvenient?
2. What do people currently do? Do they use spreadsheets, notes apps, calculators, or other websites to handle it?
3. What could I make easier? Could a simple website automate a calculation, organize information, or provide a useful recommendation?

For example:

- Problem: I never know how much time I actually spend studying.
- Current solution: I guess, or try to remember how long I worked.
- Website idea: A study tracker with a timer, a log of completed sessions, and a graph of weekly study hours.

Notice how that idea already gives you a clear plan for what to build.

Step 3: Brainstorm ideas that are realistic for your class

Since you're building with Drafter, a Python frontend library, you can make interactive websites that accept information, process it with Python, and display useful results.

Here are six ideas I'd seriously consider.

1. 💸 Student Budget Calculator

The goal: Help students understand and manage their spending.

Possible features:

- Enter purchases and their prices.
- Categorize spending, such as food, transportation, and entertainment.
- Calculate total spending in each category.
- Display a bar chart showing where the money goes.

Why it's a good project: You can practice lists, dictionaries, calculations, and graphs. You

could also save purchases in the browser so they remain available after refreshing the page.

2. 📚 Study Session Tracker

The goal: Help students understand their study habits.

Possible features:

- Start and stop a study timer.
- Record the subject studied and the time spent.
- Calculate total study hours.
- Display a graph comparing study time across subjects.

Why it's a good project: It has a clear purpose, and you can start with a simple timer before adding data tracking and visualization.

3. 🍽️ What Should I Eat?

The goal: Suggest meals based on ingredients a user already has.

Possible features:

- Select ingredients from a list.
- Specify dietary preferences.
- Match the selected ingredients to a small collection of recipes.
- Display the best matches.

Why it's a good project: It involves processing and comparing data. You can create your own small recipe collection rather than needing a database or AI-powered recommendation system.

4. 🏋️ Workout Progress Tracker

The goal: Help users track their exercise progress.

Possible features:

- Record workouts and exercises.
- Enter sets, repetitions, or weights.
- Calculate weekly workout totals.

- Graph progress for a selected exercise.

Why it's a good project: It combines user input, data storage, calculations, and visualization.

5. 🎯 Random Activity Generator

The goal: Help people decide what to do when they're bored.

Possible features:

- Select a category, such as indoor activities, outdoor activities, or social activities.
- Specify how much time is available.
- Generate a random suggestion that fits those constraints.
- Save favorite suggestions.

Why it's a good project: It's relatively easy to build, but you can make it more interesting with filtering, random selection, and personalized options.

6. 🗺️ Local Adventure Planner

The goal: Help users find ideas for things to do in a particular area.

Possible features:

- Select an activity type.
- Set a budget or distance preference.
- Display a list of matching places from a predefined collection.
- Show locations on a map.

Why it's a good project: It could have an impressive visual result. However, live location data, map APIs, and external services add complexity, so I'd start with a small predefined dataset rather than trying to build a full Google Maps alternative.

Step 4: Think about what makes a good CS1 project

Before choosing an idea, make sure it fits the technology and your time limit.

Drafter runs Python in the browser using Pyodide, so you can perform calculations, manage

data, and create interactive components without building a backend server or database.

That means your project should ideally have these three parts:

1. Input: The user enters information or makes selections.
2. Processing: Your Python program calculates something, filters data, or makes a decision.
3. Output: The website presents a result, recommendation, chart, or summary.

For example, a budget tracker could work like this:

Stage	What happens
Input	The user enters purchases and prices.
Processing	Python calculates total spending by category.

Output	The website displays totals and a chart.

That's already a complete programming project!

Keep the scope manageable

I'd recommend choosing:

- One main problem.
- Three or four core features.
- A small amount of data to manage.
- A visual element, such as a graph, if it makes sense.
- Browser-based storage if you need information to persist between visits.

Avoid making your first version depend on user accounts, a custom backend, real-time collaboration, or complicated external APIs. Those features introduce problems you don't need to solve for a successful introductory project.

Step 5: Use brainstorming to narrow your idea

You don't have to pick an idea immediately. Let's figure out what kind of website you'd actually enjoy building.

Here are three different ways we can brainstorm:

A. Practical: A website that solves a real problem for college students, such as managing assignments, spending, or studying.

B. Creative: A website centered on hobbies, entertainment, games, recommendations, or interactive experiences.

C. Technical: A website that uses graphs, timers, maps, calculations, or data processing to create something interesting.

We can take whichever direction appeals to you, generate several ideas, compare their difficulty, and pick one that fits your three-week project.

Which direction sounds most interesting: A (practical), B (creative), or C (technical)?

i want to make a website that converts binary into decimal numbers or hexadecimal into decimal or binary

I think this is a great idea for your CS1 final project! A number system converter is useful for computer science students, has a clear purpose, and is realistic to build in three weeks using Drafter and Python.

The key is to make it more than just a calculator: you could build an interactive conversion tool that shows users the results and helps them understand how number systems work.

1. Your Website Idea: Number System Converter

Your website would allow users to convert numbers between three number systems:

- Binary (base 2): Uses only `0` and `1`.

- Decimal (base 10): The standard number system we use every day.
- Hexadecimal (base 16): Uses `0-9` and `A-F`.

For example, a user could enter the binary number `1010`, and your website would display:

Number system	Result
Binary	`1010`
Decimal	`10`
Hexadecimal	`A`

You could let users choose which number system they're starting with and which one they want to convert to.

2. Features I Would Include

I'd divide the project into three levels so you can get a working version finished before adding extra features.

Level 1: Core Features (Must Have)

These features would make the website functional.

1. Number input

- Provide a text box where the user enters a number.
- Let the user choose the input base: binary, decimal, or hexadecimal.
- Let the user choose the output base.

2. Conversion

- Convert the input into the selected number system.
- Display the converted result clearly.

3. Input validation

- Prevent invalid binary numbers, such as `1021`.
- Prevent invalid hexadecimal numbers, such as `12G`.
- Display an error message when the input is invalid.

4. Clear button

- Allow the user to reset the input and output.

Level 2: Features That Make It Stand Out

Once the basic converter works, consider adding these features.

1. Convert to all three systems simultaneously

Instead of making users select an output base, display all three results at once.

For example, entering hexadecimal `2F` would display:

Number system	Result
Binary	`101111`
Decimal	`47`

Hexadecimal `2F`

This would make the website faster and more useful.

2. Show the conversion steps

For example, converting binary `1010` into decimal:

$$10102=(1\times23)+(0\times22)+(1\times21)+(0\times20)$$

1010

2

=(1×2

3

)+(0×2

2

)+(1×2

1

)+(0×2

0

)

=8+0+2+0=10

=8+0+2+0=10

Your website could display these steps to help students understand how the conversion works.

3. Add a copy button

Allow users to copy the converted number to their clipboard.

Level 3: Optional Features

Only consider these after the core features work.

- Conversion history: Show previously converted numbers.
- Dark mode: Give users a choice between light and dark themes.
- Practice mode: Generate conversion questions and let users check their answers.
- Bit visualization: Display binary digits as individual boxes, highlighting the powers of two they represent.

My recommendation is to implement the first two levels and then choose one optional feature.

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3. How You Would Build It with Python

The good news is that Python already has built-in functions for converting between number

systems.

Here are the functions you would use:

Function	Purpose	Example
`int(value, 2)`	Convert binary to decimal	`int("1010", 2)` → `10`
`int(value, 16)`	Convert hexadecimal to decimal	`int("2F", 16)` → `47`
`bin(value)`	Convert decimal to binary	`bin(10)` → `'0b1010'`
`hex(value)`	Convert decimal to hexadecimal	`hex(47)` → `'0x2f'`

You don't need to implement the mathematical conversion algorithms from scratch unless your instructor requires it.

For a converter that displays all three systems, the basic logic is:

1. Read the user's input and selected number system.
2. Convert the input into a decimal integer using Python's `int()` function.
3. Convert that integer into binary using `bin()`.
4. Convert it into hexadecimal using `hex()`.
5. Remove the `0b` and `0x` prefixes if you want to display just the digits.
6. Display the results on the website.

Important: Your program should validate the input and handle errors. For example, `int("102", 2)` raises a `ValueError` because `2` is not a valid binary digit.

These functions are compatible with Python running in Drafter's browser environment, so you won't need a backend or database.

4. Let's Connect This to Your Project Worksheet

Your assignment asks you to identify a problem, describe your solution, and explain the input, processing, and output of your program.

Here's how your idea fits those requirements.

Part 1: The Problem and Solution

Problem: Students learning computer science often need to convert numbers between binary, decimal, and hexadecimal. Performing these conversions manually can be time-consuming and lead to mistakes.

Prior solutions: Programmable calculators, online number base converters, and programmer calculators already provide number conversion functionality.

Your solution: A web-based number system converter that converts binary, decimal, and hexadecimal numbers, validates input, and optionally explains the conversion process.

Part 2: Input-Process-Output (IPO)

Component	Description
Input	A number entered by the user and the number system it belongs to.
Processing	Validate the input, convert it to a decimal integer, and convert that integer to number system or systems.
Output	The converted number or numbers, along with any applicable conversion st messages.

The core goal in one sentence:

Create a web application that converts numbers between binary, decimal, and hexadecimal while helping users understand the conversion process.

Part 3: Data Abstraction

Your program doesn't need complicated data structures to represent the problem.

Core things (nouns):

- Number
- Number system
- Conversion result
- Conversion history, if you implement that feature

Properties you need to track:

For a number, you might track:

- The entered value.
- Its original number system.
- Its decimal equivalent.
- Its converted representation.

You can ignore properties such as who entered the number, when it was originally created, or where it came from.

Part 4: Challenges and Goals

Potential challenges:

- Validating that numbers contain only digits permitted by their base.
- Handling invalid input without crashing the program.
- Displaying hexadecimal letters consistently.
- Designing a clear interface that makes conversions easy to understand.

How you'll use AI: Ask for explanations of individual functions and code sections, test the code with known conversions, and make sure you understand how the input, processing, and output connect.

Your goal: Build a functional, visually appealing number system converter that could be useful to other students learning computer science.

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5. My Recommendation for Your Final Project

I'd build a website called BaseBuddy or Base Converter with the following layout:

- A title and short explanation.
- An input box for entering a number.
- A dropdown to select Binary, Decimal, or Hexadecimal.
- A prominent Convert button.
- Three result cards displaying Binary, Decimal, and Hexadecimal.
- An optional expandable section showing the mathematical conversion steps.

- An error message for invalid input.

Keep the design clean and make the results easy to read.

The most important thing is to get the converter working first. Once that works reliably, adding conversion explanations or a practice mode would give the project more depth without making it unnecessarily complicated.

For building that binary/hex-to-decimal converter, hosting an HTML site is one option.

Ads do not influence the answers you get from ChatGPT. Your chats stay private. Learn about ads and personalization

How can i counter the already existing solutions to my website

The best way to compete with existing number converters is not to build a converter that does everything they already do, but to make yours better for a specific audience or purpose.

Since your website converts binary, decimal, and hexadecimal, I'd focus on making it especially useful for CS students who are learning number systems, rather than just people who need a quick conversion.

Here are some ways you could differentiate your website.

1. Identify what existing websites already do

Your competitors include online number base converters and programmer calculators.

Existing solution	What it does well	Opportunity for you
Standard online converters	Quickly convert between number bases.	Explain how the conversion

Programmer calculators	Provide multiple number systems and other programming-related calculations.	Offer a simpler interface des beginners.
Educational websites	Explain the mathematics behind number systems.	Combine explanations and i conversions in one place.
Manual calculations	Help students practice the conversion process.	Let students check their ans identify mistakes.

You don't need to claim these existing solutions are bad. Instead, identify a particular need they may not fully address.

2. Five ways to make your website different

Idea 1: Show the conversion steps

My top recommendation.

Most users who need a quick answer only need the result. But students learning binary and hexadecimal often need to understand *why* the result is correct.

For example, a user enters `1011` in binary.

Your website displays:

Step 1: Convert binary to decimal

$$1011_2 = (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0)$$

1011

2

=(1×2

3

)+(0×2

2

)+(1×2

1

)+(1×2

0

)

Step 2: Calculate the result

$$8+0+2+1=11$$

$$8+0+2+1=11$$

Final answer:

- Binary: `1011`
- Decimal: `11`
- Hexadecimal: `B`

Your competitive advantage: Users can learn from the conversion instead of simply receiving an answer.

This is also realistic for a CS1 project because you can generate the explanation using Python.

Idea 2: Add a learning or practice mode

Instead of only converting numbers, allow students to test their understanding.

For example:

Practice question: Convert `1101` from binary to decimal.

The user enters `13` and clicks Check Answer.

Your website responds with:

- Correct! `1101` in binary equals `13` in decimal.
- Or: Not quite! Here's how to calculate the correct answer.

You could also generate random questions and keep track of the user's score.

Your competitive advantage: Your website becomes a learning tool rather than just a calculator.

Idea 3: Make the interface beginner-friendly

Some programming tools have interfaces with many features that beginners don't need.

You could create a simple interface with three result cards:

Binary	Decimal	Hexadeci
`1011`	`11`	`B`

Whenever a user enters a valid number, the website displays all three representations.

You could also include:

- Clear error messages.
- A copy button for each result.
- Short explanations of binary, decimal, and hexadecimal.

- A clean layout that works on different screen sizes.

Your competitive advantage: The website is easy to understand and use, even for someone encountering number systems for the first time.

Idea 4: Visually represent binary numbers

This feature could make your website more memorable.

Suppose a user enters `1011`.

You could display the binary digits as individual boxes:

1 0 1

8 0 2 1

Each digit is paired with its corresponding power of two.

The user can see that:

$$8+0+2+1=11$$

$$8+0+2+1=11$$

You could highlight the positions containing `1` to show which powers of two contribute to the result.

Your competitive advantage: Users can visualize how binary numbers work instead of

having to interpret a mathematical formula alone.

Idea 5: Help users understand hexadecimal

Hexadecimal can be confusing because it uses letters alongside digits.

You could add a small reference chart:

Decimal	Binary	Hexadeci
10	1010	A
11	1011	B

12	1100	C
13	1101	D
14	1110	E
15	1111	F

When a user converts a hexadecimal number, your website could explain how each hexadecimal digit corresponds to four binary bits.

Your competitive advantage: The converter also acts as a reference tool for students studying computer science.

3. Which features should you actually build?

Because this is a three-week introductory programming project, I wouldn't attempt all five ideas.

I'd prioritize them like this:

Feature	Difficulty	P
Convert between binary, decimal, and hexadecimal	Easy	Essential
Display all three results simultaneously	Easy	Essential

Validate invalid input	Easy	Essential
Show the mathematical conversion steps	Moderate	High
Display a visual representation of binary digits	Moderate	High
Add a practice mode with scoring	Moderate	Optional
Track user accounts and online leaderboards	Difficult	Avoid

You can build the essential features using Python and Drafter without needing a backend or database.

4. How to write this in your project worksheet

Your assignment asks you to identify prior solutions and describe your own solution.

You could use the following as a starting point for your response.

Prior Solutions:

Existing online number system converters allow users to convert numbers between binary, decimal, and hexadecimal. Programmer calculators also provide number base conversions. However, these tools may prioritize obtaining a quick result over helping beginners understand the conversion process.

My Solution:

My website will convert numbers between binary, decimal, and hexadecimal while focusing

on students learning computer science. In addition to displaying the converted results, it will explain the mathematical steps involved in the conversion. I also plan to include a visual representation of binary digits to help users understand how powers of two contribute to a decimal number.

Competitive Advantage:

My website will differentiate itself by combining number conversion with educational explanations and visual learning. Rather than simply providing an answer, it will help students understand how number systems work. The interface will be simple and beginner-friendly, making it useful as both a conversion tool and a study resource.

One important note: the claims about what existing tools emphasize are hypotheses, not verified findings. Before submitting your worksheet, compare two or three actual converters and record which features they offer. That will help you identify a genuine gap rather than assuming a competitor lacks a feature.

My final recommendation

Build an educational number system converter with three defining features:

1. Instant conversion: Display binary, decimal, and hexadecimal results together.
2. Step-by-step explanations: Show users how the conversion is calculated.
3. Visual learning: Illustrate the relationship between binary digits and powers of two.

That gives your project a clear identity: a converter designed not just to give students the answer, but to teach them how to get it.

what is the most complex and over ambitioud feature meantioned and how could i make it more simpler

For your number system converter website, I think the most complex and potentially overambitious feature is the interactive practice mode with randomly generated questions, answer checking, and score tracking. Another potentially ambitious feature is a detailed visual representation of binary conversions.

Since your project is only three weeks long and you're using Drafter for a CS1 course, I'd recommend simplifying the practice mode rather than removing it entirely.

1. The Most Complex Feature: Interactive Practice Mode

The original idea included:

- Generating random conversion questions.
- Allowing users to submit answers.
- Checking whether answers are correct.
- Providing explanations for incorrect answers.
- Tracking scores across multiple questions.

Why is it overambitious?

Each individual feature is manageable, but combining them requires additional programming logic. You would need to handle different number bases, validate user answers, generate questions, and maintain the user's score as they progress.

It could take time away from building and testing your website's primary conversion functionality.

How could you simplify it?

Instead of building a complete practice system, create a single-question practice feature.

Your simplified version could:

1. Display one randomly generated binary-to-decimal question.
2. Allow the user to enter an answer.
3. Check the answer and display whether it is correct.
4. Show the correct answer if the user gets it wrong.

You could leave out score tracking, multiple difficulty levels, and progress statistics.

This preserves the educational value while reducing the amount of code you need to write.

2. Another Potentially Complex Feature: Visual Binary Explanations

The original idea included displaying binary digits in individual boxes, highlighting the powers of two, and illustrating how those values combine to produce a decimal number.

This could become complicated if you tried to create animations or interactive elements for every conversion.

A simpler alternative: Display a static table explaining the conversion.

For example, converting binary `1011` to decimal:

Binary digit	1	0	1
Power of 2	8	4	2
Contribution	8	0	2

Then display:

Decimal result: 8 + 0 + 2 + 1 = 11

This is easier to implement and still helps students understand how binary conversion works.

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3. What You Could Write in Your Project Worksheet

Your assignment asks you to identify the most complex feature, explain why it is too complicated for a three-week introductory programming project, and ask for a simpler alternative.

Here is a draft you can adapt for that section.

The most complex feature I considered was an interactive practice mode that generates random conversion questions, checks user answers, provides explanations, and tracks scores over time. This feature could be too ambitious for a three-week introductory programming project because it requires additional logic for generating questions, validating answers, and maintaining scores.

A simpler alternative would be to display one randomly generated conversion question at a time and allow the user to submit an answer. The website would tell the user whether the answer is correct and display the correct answer if necessary. I would leave out score tracking and advanced progress statistics to keep the project manageable while still providing an educational feature.

My recommendation

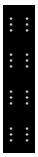
Build your website in this order:

1. Core converter: Convert between binary, decimal, and hexadecimal.
2. Input validation: Handle invalid numbers without crashing.
3. Conversion explanations: Show how binary numbers convert to decimal.
4. Optional practice question: Add a simple question-and-answer feature if time permits.

That way, you'll have a complete, functional website even if you don't finish the optional feature.

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Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.